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Report on Industrial Training

At

SRI LANKA TELECOM PLC

Colombo 01



UNIVERSITY OF COLOMBO

Name	: K.A. Hirushima
Student Number	: 2021T01120
Course	: Bachelor of Engineering Technology
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Preface

I completed a 24-week industrial training program at Sri Lanka Telecom PLC as a requirement of the Bachelor of Engineering Technology Honors degree in Instrumentation and Automation Technology at the University of Colombo. The Faculty of Technology in collaboration with the National Apprentice and Industrial Training Authority (NAITA) arranged this training

This internship gave me my first professional workplace experience, when I was able to put the theoretical knowledge I had learnt at university to use in actual workplaces. I was expose to a wide range of technical and practical fields throughout this time such as,

- Embedded systems (ESP32 programming, ULP coprocessor, Arduino IDE,
- IoT & Networking (MQTT (QoS 2), AP + STA mode, ThingsBoard devices, dashboards, rule chains),
- Testing & Calibration, documentation and Technical Communication.

There are three chapters in this report. The first chapter introduces the training company and its structure, business activities, and management practices. The second chapter describe my training experience, tasks, projects, and challenges faced during the training period. The last chapter, where I highlight the expertise and skills have acquired, as well as my and suggestions for both my personal development and for future training improvements.

The main purpose of this report is to provide a clear describe the experience, knowledge, and skills I earned throughout this training period, and how it contributed to my professional and personal growth.

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Acronyms and Abbreviations

- IoT - Internet of Things
- MQTT - Message Queuing Telemetry Transport
- OTA - Over-The-Air
- IR - Infrared
- FOTA - Firmware Over-The-Air
- IIoT - Industrial Internet of Things
- CoAP - Constrained Application Protocol
- LoRaWAN - Long Range Wide Area Network
- MCU - Microcontroller
- MPU - Microprocessor
- UART - Universal Asynchronous Receiver-Transmitter
- SPI - Serial Peripheral Interface
- I2C - Inter-Integrated Circuit
- HTTP - HyperText Transfer Protocol
- REST API - Representational State Transfer
- AP - Access Point mode.
- QoS - Quality of Service
- CT - Current Transformer
- ULP - Ultra-Low Power
- CPU - Central Processing Unit
- CLP - Constraint Logic Programming
- RTC - Real-Time Clock
- PWM - Pulse Width Modulation

1 Chapter 01

1.1 Introduction about Training Institute



Figure 1 - The Company Logo

Sri Lanka Telecom PLC (SLT) is the leading telecommunication and digital service provider in Sri Lanka. Broadband internet, digital television, fixed and mobile communication and enterprise ICT solutions are just a few of the many services it provides. Operating under the brand SLT Mobitel, the company serves millions of individual customers, companies, and government customers across the island. As the national telecommunications provider, SLT plays a vital role in the country economic expansion and digital transformation. By providing reliable connectivity and advanced communication technologies for education, healthcare, e-commerce, and government services. Its large-scale investment in fiber optic networks, mobile technology, and digital platforms has made it an essential partner in Sri Lanka's transformation to a knowledge-based and technologically advanced society.

Sri Lanka Telecom PLC (SLT), established in 1858, is Sri Lanka's oldest and leading telecommunications service provider. The corporation started out by establishing the nation's first

telegraph line between Galle and Colombo, which signaled the start of telecommunications services. The corporation started out by establishing the nation's first telegraph line between Galle and Colombo, which signaled the start of telecommunications services.

SLT has undergone significant change over the years. After being corporatized in 1991, it was privatized in 2000 through a strategic alliance with Japan's Nippon Telegraph and Telephone Corporation (NTT).Following NTT's 2008 divestment, the Government of Sri Lanka acquired 48% of the shares, while Global Telecommunications Holdings N.V. of the Netherlands purchased the remaining 44.98%.

Sri Lanka Telecom operates multiple business areas, offering products and services for modern communication, information and technology needs.

- Fixed-line and Mobile Communication Services
- Broadband Internet and Fiber Optic Services
- IPTV, Cloud Computing, and Digital Solutions
- ICT Infrastructure and Enterprise Solutions

The Research and Development (R&D) Department, also known as the Digital Lab, at Sri Lanka Telecom PLC plays a pivotal role in driving technological innovation in the country’s telecommunications sector. The creation and integration of A IoT solutions, intelligent automation systems, and cutting-edge communication platforms are just a few of the significant milestones the lab has accomplished. By promoting a culture of continuous innovation, the R&D team contributes to the creation of sustainable and forward-looking solutions that improve connectivity and digital services across Sri Lanka. These programs strengthen SLT's position as a leader in innovation and technology while also improving the lives of its residents and assisting the country's transition to a digitally empowered society.

1.1.1 Organizational Structure

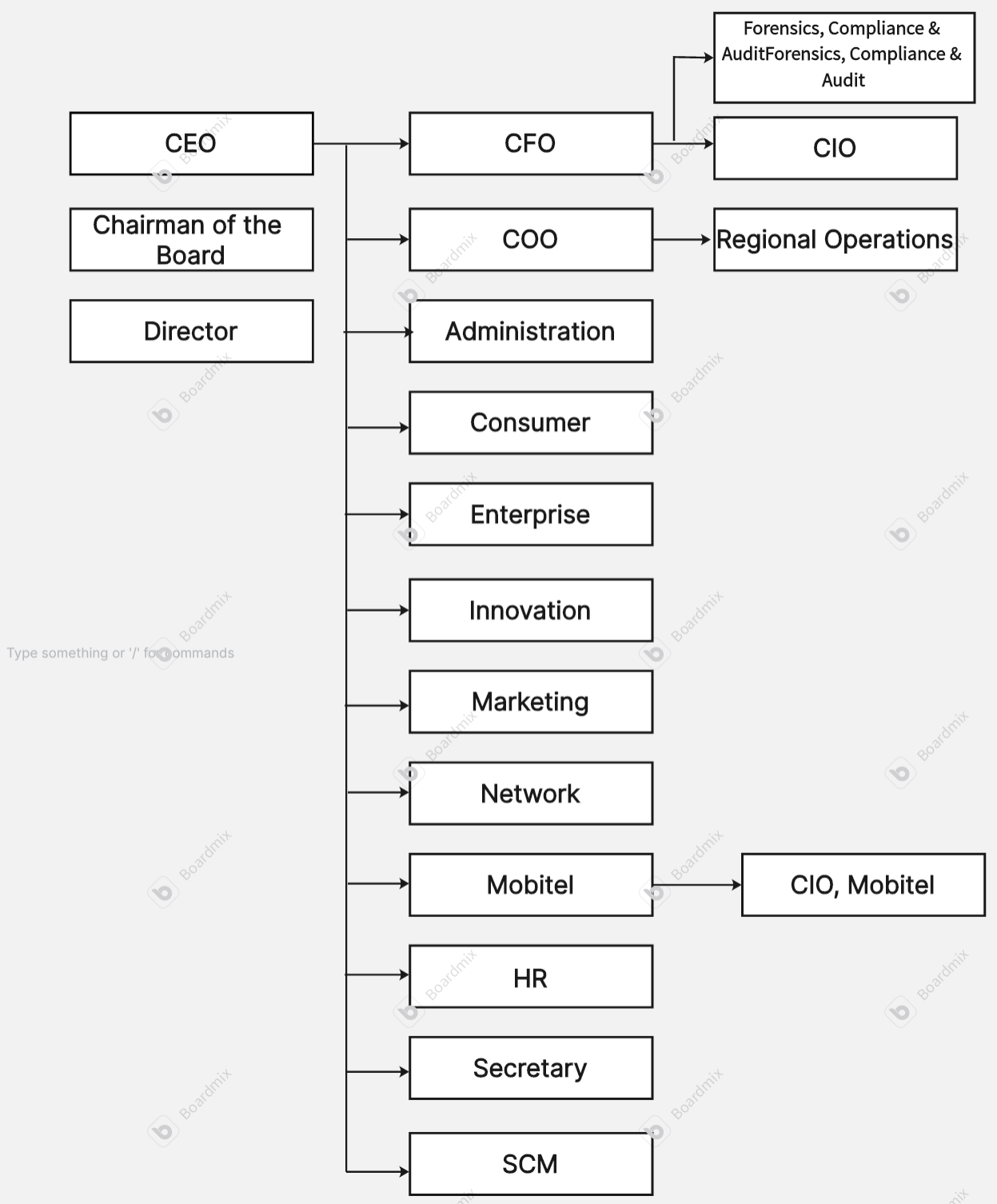


Figure 2 - SLT Organizational Chart

1.1.2 Vision of the Company

"All Sri Lankans seamlessly connected with world-class information, communication and entertainment services."

1.1.3 Mission of the Company

"You're trusted and proven partner for innovative and exciting communication experiences delivered with passion, quality and commitment"

1.1.4 Values of the Company

- Trustworthy – SLT keep promises and build trust.
- Innovative – SLT use new ideas and technology to improve services.
- Responsive – SLT pay attention to people's needs and respond promptly.
- Teamwork – SLT Employees work together to reach goals.
- Excellence – SLT aim to do every task at the highest quality.
- Results Driven – SLT focus on success and growth for the company and shareholders.

1.1.5 Nature of the business operations and Main Function

Sri Lanka Telecom is the largest telecommunication company in Sri Lanka. It provides many services for individual users, businesses, covering communication, internet and digital solution Fixed-line and Mobile Communication Services. Fixed-line and Mobile Communication Services - SLT provides traditional fixed-line telephone connections and advanced mobile communication services. These services include voice calls, SMS, and modern mobile applications, enabling reliable communication for people across the country. They also provide Broadband Internet and Fiber Optic Services - They offers fast internet through ADSL, 4G/5G, fiber optics connection. Its fiber optic network given stable and high-speed internet. Further, provide clouded services such as PeoTV, a digital TV service with interactive features. Also offers cloud computing and digital platforms its helping businesses for store data, host applications, and manage IT solutions. And they are provide ICT and Enterprise Solutions.

1.1.6 Employees

As a December 31, 2024, Sri Lanka Telecom has approximately 8,050 individual employees. SLT has multiple departments and variety of job roles such as, Technical and Engineering Staff are responsible for maintain and upgrading SLT network infrastructure and ensuring stable communication services across the country. Another one is Management and Administrative Personnel they are created strategic planning and corporate governance, playing a crucial role in steering the company's direction. Support and Customer team also playing important role in SLT they are handle customer inquiries, provide technical support, and manage service-related issues, ensuring a positive customer experience.

1.2 Management of Sri Lanka Telecom

1.2.1 Leave Policies

SLT provides different type of leave for help employees to work and life balance and maintain they are health and personal or family matters.

- 14 days of annual leave
- Casual leave of up to 7 days per year
- Sick leaves
- Maternity leave entitles female employees to 84 working days
- Paternity leave to fathers for a few days.

Unless there is an urgent or medical reason, requests for leave or work from home must be sent to Sri Lanka Telecom (SLT) three days before the planned day. In such cases, it is preferable to inform management before the start of the workday, unless the employee is unable to do so. The same conditions apply to extensions of leave or work from home. Failure to comply with these requirements considered uninformed leave. For interns, uninformed leaves recorded as unauthorized

1.2.2 EPF (Employees Provident Fund)

SLT employees contribute 8% of their basic salary while the company contributes 12%. This makes a total contribution of 20%. Which is save in the employees EPF account. The fund helps employees after retirement, in case of disability, or supports their family if they pass away.

1.2.3 ETF (Employees Trust Fund)

SLT also contributes 3% of the employee basic salary to the ETF. Managed by the Employees Trust Fund Board under the Ministry of Finance. This provides additional financial support, welfare benefits, and insurance coverage for employees.

1.2.4 Recruitment Procedures

At Sri Lanka Telecom, the Human Resources (HR) department together with the relevant management teams carefully manages the recruitment process. The main goal is to hire the most suitable people for the company needs. Recruitment usually takes place when SLT launches new services or when employees retire, resign, or are promote.

The following are the primary steps in the hiring process:

- Finding out how many employees are needed
- Advertising job openings
- Checking applications and shortlisting candidates
- Interviewing the selected applicants
- Offering jobs to successful candidates

For interns or entry-level employees, SLT provides a three-month training program followed by a six-month probation period. After this, permanent employment given if their performance is good.

This approach supports SLT's expansion in the telecom sector by assisting the company in retaining strong and experienced personnel.

1.2.5 Welfare Activities in SLT

Sri Lanka Telecom provides many of welfare activities to ensure employee well-being and motivation. These include medical and health benefits, insurance schemes, recreational facilities, housing loans, transport services, training programs, and social events. SLT also promotes work-life balance by supporting sports, cultural activities, and family assistance program

2 Chapter 02

2.1 Training Experience at Digital Lab Research and Development

The internship period marks an important stage in academic and professional development. This training provides direct exposure to real industry practices and creates opportunities to apply theoretical knowledge in practical situations. The work focused on areas such as embedded systems, automated procedures, software testing, documentation, and project management. Special attention was given to practical tasks including system design, data acquisition, hardware integration, and troubleshooting, which helped to connect academic knowledge with real-world applications. . Communication and organizational skills were further enhanced by taking part in team talks, technical problem-solving exercises, and quality assurance exercises. The internship also develops professional discipline, time management, and organizational understanding. This valuable experience builds a strong foundation for future career goals and helps create a clear link between education and workplace requirements.

2.2 Introduction to IoT and Embedded Systems

2.2.1 What is IoT?

The Internet of Things (IoT) is a network of smart devices such as fitness trackers, thermostats, appliances, and even light bulbs that connect to the internet, collect data, and communicate with each other. With the help of sensors and software, these gadgets enhance the intelligence and

automation of commonplace systems, ranging from industrial machinery and smart cities to smart homes and wearable health monitors. However, while IoT offers efficiency and innovation, it also raises concerns about privacy and security. Many devices released without strong protections, creating risks of hacking, data misuse, or large-scale cyberattacks. IoT brings both opportunities and responsibilities.

2.2.1.1 Role of IoT in Modern Applications

The idea of linking commonplace things to the internet in order enable exchange of information and intelligent decision-making known as the Internet of Things (IoT). IoT applications span across multiple domains

- Smart Homes: smart thermostats, automated lighting and security systems.
- Transportation: Vehicle tracking, predictive maintenance, and traffic management.
- Industry (IIoT): energy management, Factory automation and predictive maintenance.
- Healthcare: Wearable devices for monitoring heart rate, glucose levels, and patient activity.

2.2.2 What are Embedded Systems?

A hardware and software combination created to carry out a particular task is called an embedded system. In contrast to general-purpose computers (such as laptops), embedded systems are designed to perform specific functions like operating a washing machine, keeping an eye on a patient's pulse rate, or controlling the brake system of a car. Typically, they are compact, effective, and designed to function best in real-time scenarios.

2.2.2.1 Features of Embedded Systems

- Low Power Consumption - Embedded systems made to use small current. Embedded system can run on small batteries for a long time.
- Real-Time Performance - They respond quickly when needed.
- Reliability - They designed to function flawlessly at all times, particularly in crucial products like medical equipment.
- Compact and Cost-Effective - They are small and not expensive to make.

2.2.3 Relationship between Embedded Systems and IoT

IoT and embedded systems combine to produce intelligent, networked applications. Embedded systems are small computing devices, such as microcontrollers, sensors, and actuators, designed to perform specific tasks. They control hardware, process data, and perform real-time operations. These embedded systems may send and receive data, communicate with other systems, and be remotely monitored thanks to the Internet of Things, which links them to a local network or the internet. While embedded systems provide real-time data and handle local processing, IoT platforms collect this data, store it, visualize it, and help analyze it for decision-making. Solutions are made possible by this combination in a number of domains, including as households, industries, healthcare, and agriculture.

2.2.4 Architecture of Embedded Systems and IoT

Embedded IoT systems are like a group, with each member has a specific task. They work together to sense the environment, transmit information and data, make decisions, and show actions. The system can be separated into multiple layers

2.2.4.1 Hardware Layer

This layer is the foundation of the system.

- Microcontrollers (MCUs): ESP32, Arduino, STM32. They read data from sensors and control actuators. These are like small computers.
- Sensors: They measure things like motion, light, temperature, and humidity. Without sensors, the system cannot understand what is going on in the environment without sensors.
- Actuators: Device that respond to command like turning on a motor, switching on a light, or opening a valve.

2.2.4.2 Communication Layer

This layer sent data between devices, microcontrollers, and the cloud. Depending on power, speed, and distance, it uses multiple methods.

- Wireless Protocols: Wi-Fi, bluetooth, LoRa and NB-IoT. Wifi for local networks, Bluetooth for short range, LoRa for long distance, and NB-IoT for cellular networks.
- Messaging Protocols: MQTT and CoAP effectively transmit little data packets.
- Serial Communication: SPI, I2C, UART connect sensors and microcontrollers directly.

2.2.4.3 Processing Layer

To make raw data useful, this layer processes it. It may occur on a nearby server or on the device itself (edge processing).

- Data Filtering: Remove noise or incorrect readings from sensors.
- Local Decisions: The system doesn't need to wait for the cloud to take action.
- Data Compression: Reduce data size before transferring to the cloud to save bandwidth.

2.2.4.4 Cloud Layer

This layer does long-term data storage and analysis. Cloud platforms like ThingsBoard, AWS IoT, or Azure IoT manage the data and give control options.

- Data Storage: save sensor reading for future uses.
- Data Analysis: Find trends or patterns, like how variations in greenhouse temperature.
- Remote Control: From any location, users can switch devices on or off.

2.2.4.5 Application Layer

The interface between the system and people is located at this layer. It presents data in an intelligible manner.

- Dashboards: Display readings, charts, or graphs.
- Alerts and notifications: Inform users in the event that something goes wrong.
- Control interfaces: Let users change things like lighting and temperature.

2.2.5 How Embedded IoT Systems Work

Sensors in an embedded Internet of Things system collect data from the real world, such as motion, light, and temperature. After processing this input data, the microcontroller generates actions.

Information is cleaned and processed by the processing layer to help in decision-making. Data storage, remote access, and visualization platforms like ThingsBoard are all made possible by the cloud. Users can monitor data, receive alerts, and operate devices via the application layer. These layers work together to give the IoT system

2.3 IoT Cloud Platforms

IoT cloud platforms work like a middle layer between devices and people. The devices, such as sensors or controllers, collect data from the real world. Through the internet, this data is sent to the cloud. After that the platform then stores the data safely and makes it easy to access from anywhere. Users can log in and view the data on dashboards, graphs, or reports. Additionally they can also send commands back to the devices, like turning a motor on or off. These platforms manage large tasks, assist manage several devices at once, and use analytics to make systems smarter. Examples are ThingsBoard, AWS IoT, and Microsoft Azure IoT.

1. Role of IoT Cloud Platforms

- They connect devices, sensors, and users.
- They enable the connection between the hardware and the internet.
- They make remote monitoring and control.

2. Data Handling

- Devices send data to the cloud.
- It is safely stored on the platform.
- The data is always available, so users can check it anytime.

3. User Interaction

- Dashboards display real-time sensor data in charts or graphs.
- Devices can be remotely controlled by users (e.g., lights, pumps).
- Notifications or alerts can let users know if something is a miss.

4. Analytics and Decision-Making

- Platforms process the data.
- This helps users make smart decisions quickly.

5. Examples of IoT Cloud Platforms

- ThingsBoard: Free and open-source, good for learning and projects.
- AWS IoT: Scalable, used in industries.
- Microsoft Azure IoT: Strong integration with Microsoft services.

2.3.1 ThingsBoard for IoT Solutions

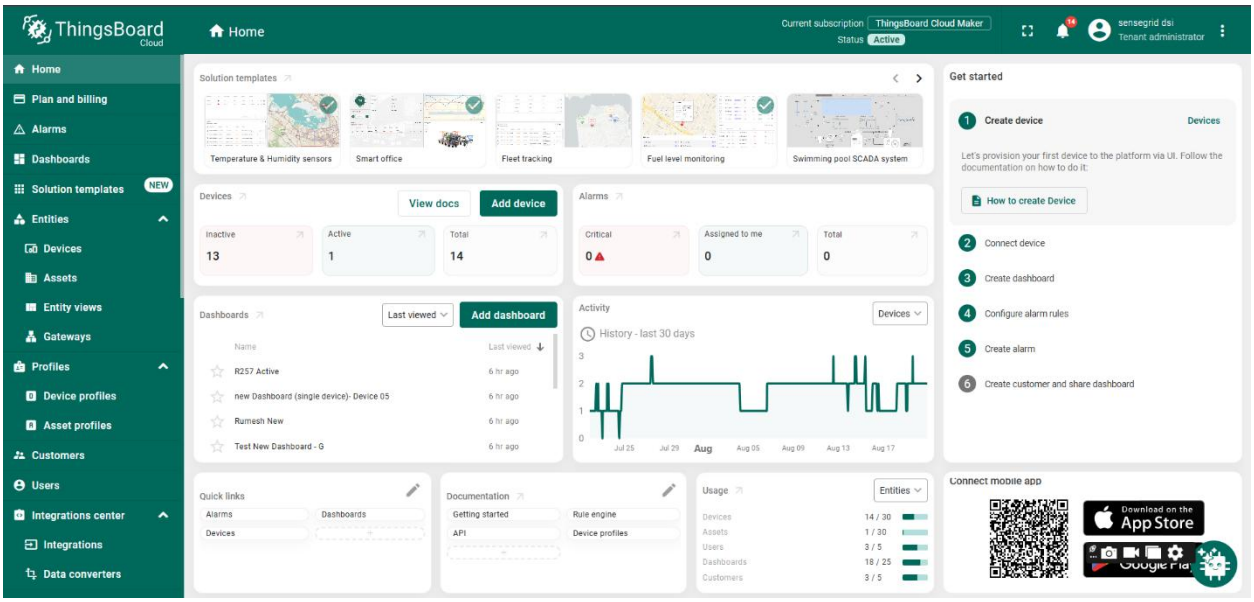


Figure 4 - Thingsboard Dashboard Cloud

A free and open-source IoT platform that works with both small and large applications is called ThingsBoard. It enables rule-based automation, real-time data visualisation, and simple device connectivity. ThingsBoard is adaptable, offers dashboards that can be tailored for various applications, and supports the MQTT and HTTP protocols. This makes it a good option for applications including environmental sensing, industrial monitoring, and smart homes.

- Data Collection: Sensors take measurements of variables like light, humidity, and temperature. Protocols like MQTT or HTTP are used to transfer the data from the device to ThingsBoard.
- Data Visualization: ThingsBoard makes it simple to track trends and spot problems by transforming raw data into graphs, charts, and gauges.
- Device Management: Users can register devices, monitor their status, configure settings, and send commands remotely. This helps manage multiple devices efficiently from one platform.

2.3.1.1 ThingsBoard Architecture

2.3.1.1.1 Device Layer

Data from IoT devices (such as ESP32, sensors, or smart machines) is sent here. Devices communicate with ThingsBoard using the supported protocols like HTTP,MQTT and CoAP for communicate between thingsboard and device .Data from IoT devices, such as ESP32, sensors, or smart machines is sent here. An access token is used to securely connect each device.

2.3.1.1.2 Transport Layer

ThingsBoard has a transport service that receives data from devices. This layer makes sure data reaches the right part of the platform.

2.3.1.1.3 Rule Engine Layer

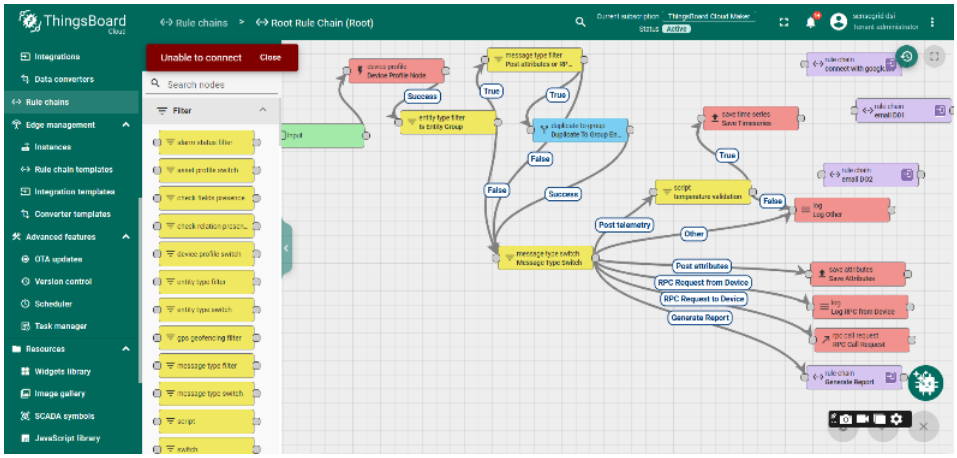


Figure 5 - Root Rule Chain

This is the “brain” of ThingsBoard. the rule engine processes incoming data, checks conditions, and decides actions.

2.3.1.1.4 Storage Layer

ThingsBoard stores device information and telemetry (time-series data) in databases. It stores data using either Cassandra or PostgreSQL by default. Data is securely kept and is retrievable for visualisation at a later time.

2.3.1.1.5 Application Layer

Dashboards, widgets, and APIs are all included in this. Clients can operate equipment, generate reports, and view data in real time. Charts, maps, alarms, and personalised visualisations are all possible with dashboards.

2.3.1.1.6 Integration Layer

ThingsBoard can connect with cloud services (AWS, Azure, Google Cloud) and external applications using REST APIs or gRPC. This helps build larger IoT solutions.

2.3.1.2 ThingsBoard supports protocols

2.3.1.2.1 MQTT (Message Queuing Telemetry Transport)

MQTT is a lightweight, straightforward messaging protocol that functions well with Internet of Things systems. Sensors and other devices transmit data to a central broker, which then distributes it to other devices or applications upon request. This approach consumes very little electricity or internet traffic and is quick and dependable. For instance, data from a temperature sensor can be sent to the broker and promptly received by a mobile app.

2.3.1.2.2 CoAP (Constrained Application Protocol)

A lightweight communication protocol called CoAP (Constrained Application Protocol) was created for tiny devices with constrained power and memory. It functions similarly to HTTP but

is far more straightforward and effective. CoAP employs UDP instead of TCP, which makes it faster and more suitable for devices on low-power or constrained networks. For instance, a smart light can swiftly receive commands like "turn on" or "turn off" by using CoAP.

2.3.1.2.3 HTTP

One of the most widely used protocols on the internet is HTTP (HyperText Transfer Protocol). Numerous programs currently support it, and it is easy to grasp. Request-response communication is how devices interact with HTTP; the device makes a request, and the server responds. For instance, a weather station can use HTTP to communicate temperature data to a cloud server, which verifies receipt of the data.

2.4 Studies during My Internship

2.4.1 Develop an ESP32 system that operates in both Access simultaneously AP and STA mode

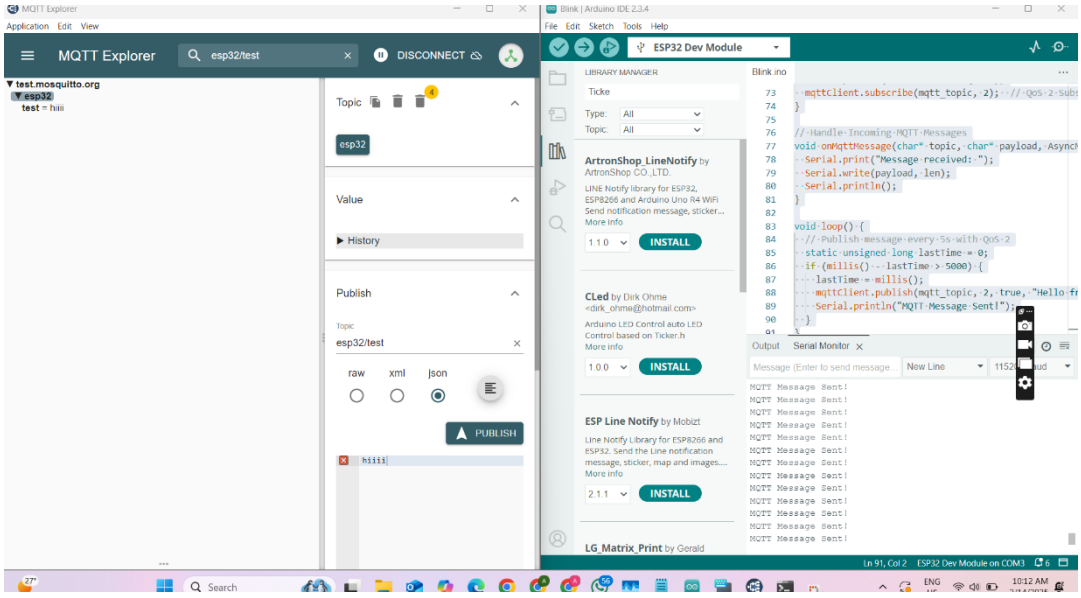


Figure 6 - MQTT Explore with Arduino IDE Framework

This project is about creating an ESP32 system that works in simultaneously both (STA) and Access Point (AP) mode. The ESP32 can often connect to Wi-Fi in three different ways. Like a laptop or cell phone, it connects to an existing router and functions as a typical Wi-Fi device when in STA mode. In AP mode, the ESP32 creates its own Wi-Fi network, and other devices can connect directly to it. The ESP32 can do both functions simultaneously in AP+STA mode: it establishes a connection to the internet through a router (STA) and simultaneously offers its own Wi-Fi hotspot (AP) for other devices to join. When we want to keep the ESP32 connected to the cloud but yet allowing direct access from nearby devices, this capability is quite helpful.

Three main libraries are used in this project, Ticker.h, AsyncMqttClient.h, and WiFi.h. STA and AP modes are configured using the WiFi.h library. Real-time message exchange between ESP32 and a MQTT broker is made possible via the AsyncMqttClient.h. specially AsyncMqttClient.h library only support MQTT QOS2 and Arduino IDE. Setting timers and repeating tasks, such sending a message every few seconds, are made easier with the Ticker.h library. First, the ESP32 is connected in STA mode to an existing Wi-Fi network, which gives it internet access. The ESP32's own Wi-Fi is then made accessible to other devices (such as a laptop or mobile phone) by turning on AP mode.

The ESP32 is linked to the public MQTT broker "test.mosquitto.org" on port 1883 in order to communicate. A topic called “esp32/test” is created, The ESP32 uses the Ticker library to publish a message to the MQTT server every five seconds. These messages can be sent quickly to other devices that are subscribed to the same subject. Other devices subscribed to the same topic can receive these messages instantly. This shows how the ESP32 can act as both a Wi-Fi client and server, while also sending and receiving MQTT messages at the same time.

2.4.1.1 MQTT

Message Queue Telemetry Transport (MQTT) is a communication protocol designed for Internet of Things (IoT) devices that function in environments with extremely high latency and limited low bandwidth. Its efficiency in low-bandwidth, high-latency situations makes MQTT an excellent choice for machine-to-machine communication. MQTT is a straightforward, lightweight messaging protocol that facilitates communication between multiple devices. As a TCP-based protocol, it employs a publish subscribe model, which is ideal for transmitting data between

resource-constrained devices with low bandwidth and power needs. This protocol is widely used in IoT frameworks because of its effectiveness and minimal overhead

2.4.1.2 QOS2

QoS 2 in MQTT refers to "exactly once delivery," which is the most dependable and secure method of message delivery. When a device transmits a message with QoS 2, when a device transmits a message using QoS 2, the system employs a unique four-step handshake between the sender and the recipient to ensure that the message is sent just once, without being lost or duplicated. First, the sender sends the message, then the receiver replies that it has received it.The sender then verifies once more, and the recipient then confirms one last time. Because of this meticulous process, QoS 2 is slower than QoS 0 (at most once) and QoS 1 (at least once), but it is particularly helpful when data correctness is crucial.

QOS2 make sure massages received exactly once. QOS2 have 4 steps

- PUBLISH – client send the massage to broker
- PUBREC – broker send the receives massage to client
- PUBREL – client conform
- PUBCOMP – broker confirm

MQTT QOS have 3 levels

- QoS 0 – At most once
- QoS 1 – At least once
- QoS 2 – Exactly once
-

2.4.2 FOTA

FOTA stands for Firmware Over-The-Air. A technology lets you update the firmware of embedded devices, IoT devices, or electronics remotely, without having to connect to the device physically. This is helpful for fixing bugs, adding new features, improving security, or changing device functionality after it has been deployed**.**

2.4.2.1 FOTA using GitHub

Firmware binary file need save as a .bin file. In addition, this file store in the GitHub repository. Device periodically checks in the GitHub new version file added. Using HTTP protocol. If new version file uploaded device compare the version, downloads, and flash it automatically. This method easy to use and this match for small project. Problem is GitHub given to limits for file size and they not built in OTA notification.

2.4.2.2 FOTA using ThingsBoard

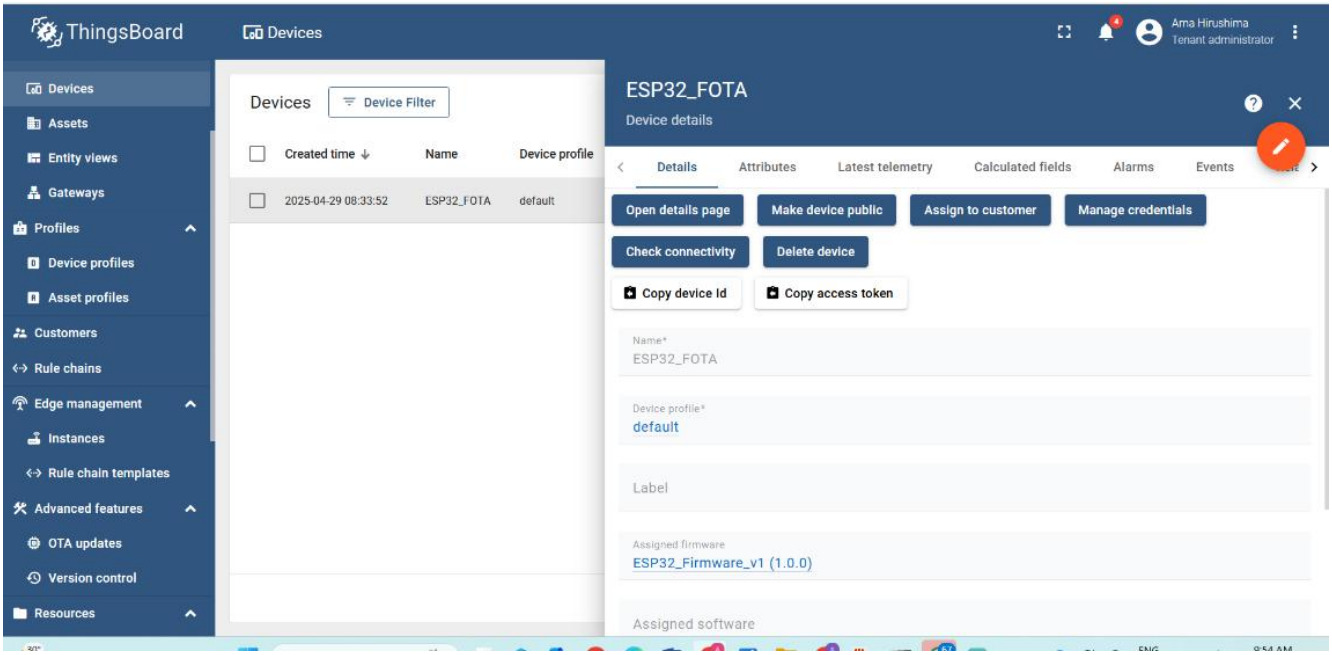


Figure 7 - FOTA setup in Thingsboard

Thingsboard is an IOT platform and can manage devices and send commands remotely. also this method need to use .bin file firmware is uploaded to as device attribute. using thingsboard can manage multiple devices and update states can monitor the real time. problem is Thingsboard has some limitations and need to pay

Steps

1. Set the things board settings for FOTA - .bin format file update firmware section

2. configuring the ESP32 for FOTA - ESP32 needs to be programmed to check for FOTA updates

- WiFi.h – wi-fi connection
- PubSubClient.h – MQTT Communication
- ArduinoJson.h – JSON data
- Update.h – Handle FOTA

- OTA Update on the ESP32 - ESP32 connected to the Things Board MQTT broker, it will periodically check for firmware updates. When new version file is available:
- The ESP32 will receive an MQTT message from Things Board with the update URL.
- The ESP32 will then download the new firmware via HTTP and perform the FOTA update.
- Once the update is complete, the ESP32 will restart with the new firmware.

2.4.2.3 ESP32 Firmware Over-The-Air (FOTA) Update Web Page

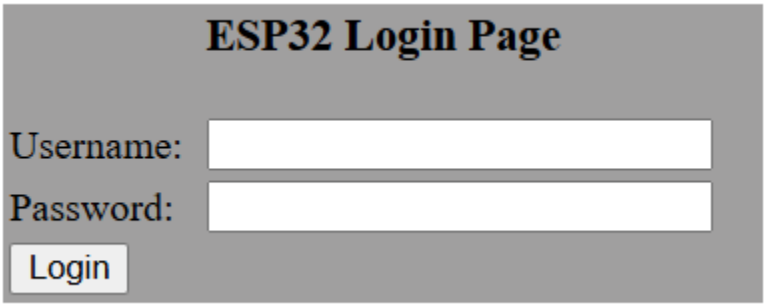


Figure 8 - FOTA Setup using webpage

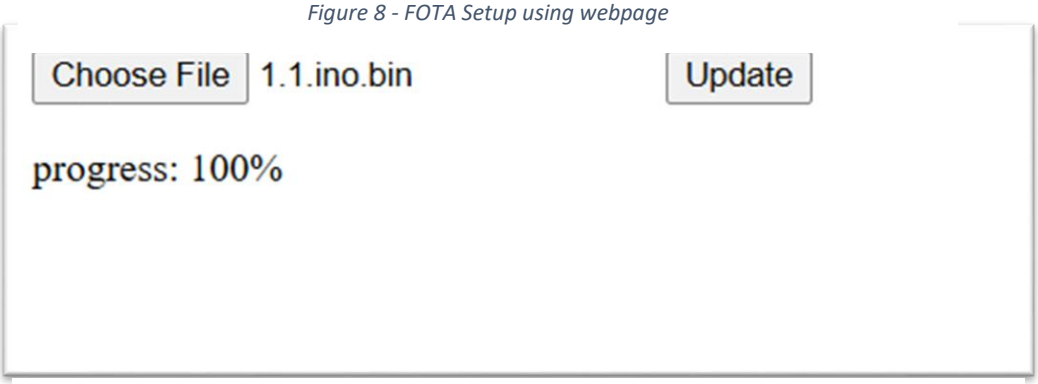


Figure 9 - Uploaded new version

This project allows to update the firmware of an ESP32 board wirelessly through a web page. When the ESP32 connects to Wi-Fi, it starts a web server that users can access from a browser. First, a simple login page appears where both the username and password must be “admin”. After logging in, the user taken to another page where they can upload a new firmware (.bin) file. The ESP32 receives this file and writes it to its memory, then restarts to run the new firmware

2.4.3 ULP Coprocessor



Figure 10 - Esp32

2.4.3.1 About ULP coprocessor

The Ultra-Low Power (ULP) coprocessor is a small helper chip inside the ESP32 that can work while the main CPU is sleeping. This helps save power and make the battery last longer. It can read sensors, control small devices, and do simple jobs without waking the main CPU. If something important happens, like a sensor reading going above a set limit, it can wake the main CPU to handle it.

CLP Coprocessors can,

- Read RTC GPIO pin - When the main CPU is, sleeping ESP32 RTC GPIO pins stay alive.
- Ultra Low power Co processor can check if pins are high or low.
- Read analog values - RTC GPIO pins can also measure voltage values and ulp can read this voltage values if CPU is sleep.
- Read touch sensors - ESP32 touch pins can sense when we touch it and ULP can monitor these touch pins in deep sleep mode.

- Do simple calculations or comparisons - ULP can only do simple task like compare voltage
- Values count pulse - Wake the big CPU if something important happens If the ULP finds that a condition is met, it can send a wake-up signal to the main CPU

ESP32 main CPU use a lot of power. The RTC domain is like a tiny battery-powered clock that still runs even if you turn off everything else. When put the ESP32 into deep sleep, main CPU, Wi-Fi, Bluetooth, peripherals are switch off and RTC timer, RTC pins, touch sensors, ULP coprocessor ON.

OFF in Deep Sleep mode

- Main CPU cores – Stop running completely; they do not execute Arduino code or loops.
- Wi-Fi & Bluetooth – Radio hardware is off, no data transmission.
- Normal GPIOs – Most pins lose input/output ability unless they are RTC pins.
- Peripherals – UART, SPI, I2C, PWM, and other hardware functions stop. ON in Deep Sleep mode
- RTC timer – Works like an alarm clock that can wake the ESP32 after a set time
- ULP coprocessor – A tiny helper processor that can monitor sensors while the main CPU sleeps
- RTC GPIO pins – Special pins that remain active to detect signals during sleep.
- Touch sensors – Can detect touches even while sleeping.
- RTC memory – Small memory area that keeps variables after waking up

2.5 Project

2.5.1 DSI Project

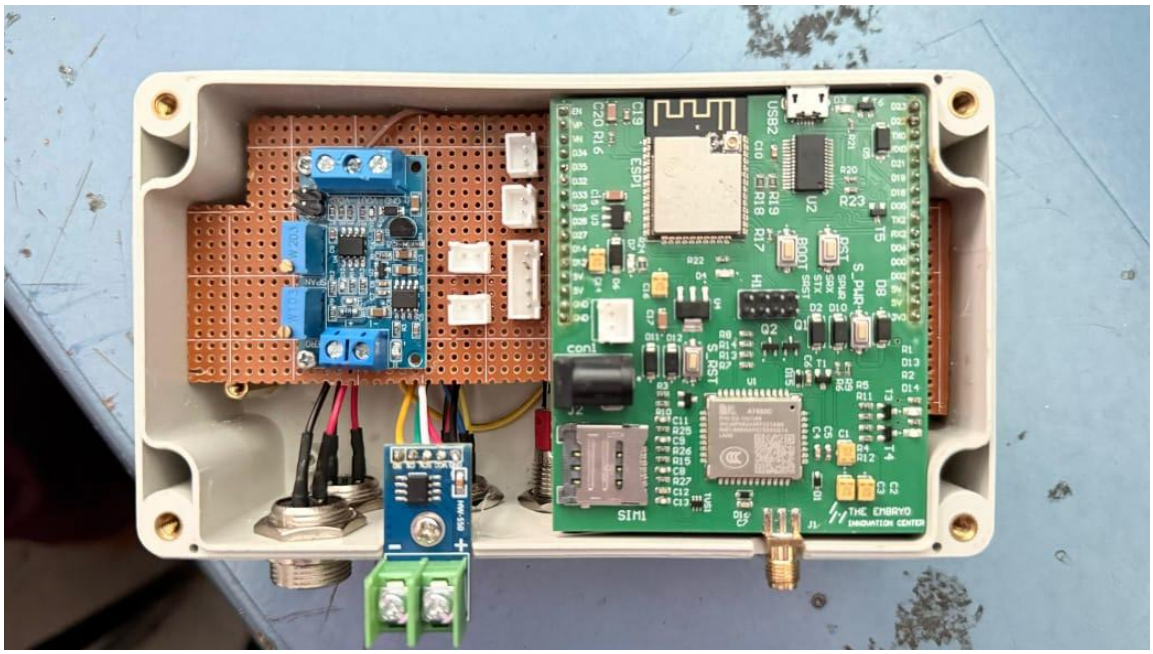


Figure 11 - One DSI device

The DSI Machine Condition Monitoring System was put in place at the DSI rubber processing plant in Galle to fix the issues that came up when machines broke down often and for long periods during workers tea and lunch breaks. Monitoring the rubber processing machine's performance and operating status in real time was the project's main objective. The technology guarantees that the business can precisely track the machine's operating condition, cut down on unnecessary idle time, and improve production efficiency by creating an online monitoring solution.

The company pointed out in the first analysis that the rubber-processing machine had extensive idle times, particularly during breaks. This downtime decreased overall productivity and efficiency since machine availability directly affects production output. In addition to continuously monitoring the machine's operating temperature, a solution was needed to automatically determine whether the machine was operating actively (load) or idlely (no-load).

System uses two current transformers and two temperature sensors for detect machine status and all sensor data is transmitted to Thingsboard dashboard using MQTT over WI-FI.

Current transformers (600A/5A) to detect if a machine is actively working

Infrared (IR) sensor (0 °C – 200 °C): Measures surface temperature without contact.

K-type thermocouple (0 °C – 85 °C): Measures internal temperature accurately

2.5.1.1 Components

2.5.1.1.1 Microcontroller

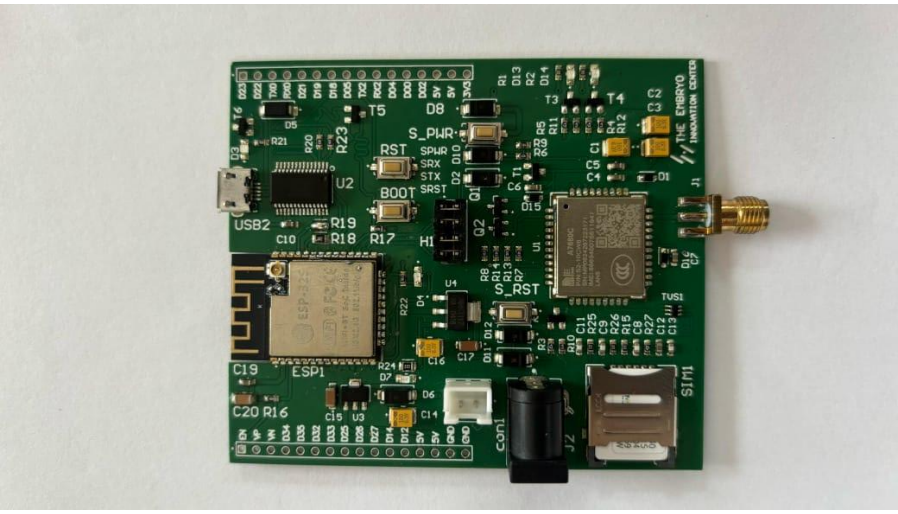


Figure 12 - SLT Cutomize Development Board

In this project, a customized development board designed by SLT was used as the microcontroller. The board is equipped with an ESP32 chip for processing and communication tasks and it includes a 4G SIM module to provide reliable internet connectivity. This setup allows the system to collect sensor data, process it locally, and then transmit it to the ThingsBoard platform for real-time monitoring and visualization.

2.5.1.1.2 Current Transformer (CT)



Figure 13 - Current Transformer

A Current Transformer (CT) is a sensor that measures AC current indirectly by producing a smaller, proportional current in its secondary coil corresponding to the primary current flowing through the conductor. In the DSI project, the CT was used to detect the machine's load status, converting a high primary current (0–600A) into a manageable output current (0–5A). Since the CT's output is still a current signal rather than a voltage signal that can be read by devices like the ESP32, it cannot be connected directly to microcontrollers.

2.5.1.1.3 ACS712

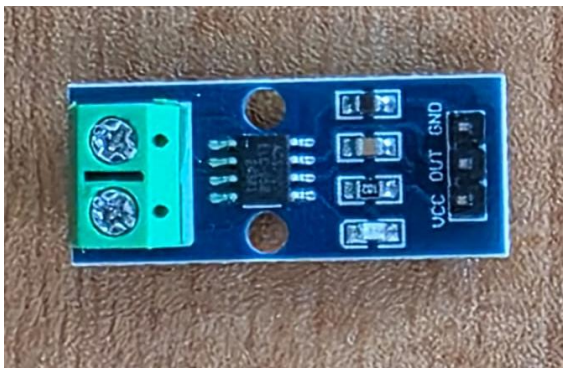


Figure 14 - ACS712 Module

The ACS712 is a current sensor that can measure both AC and DC safely. It works using the Hall-effect, which changes the magnetic field from current into a voltage signal. This voltage can be read by a microcontroller like the ESP32 for real-time monitoring. The sensor does not touch the current directly, so it is safe and protects the electronics. Its output increases in proportion to the current, making it easy to use and read. In the DSI project, the ACS712 was used with a Current Transformer (CT) to convert the CT's 0–5A output into a voltage signal for testing

Using the CT and ACS712 together is important because the CT handles high primary currents safely, and the ACS712 converts the CT's output into a voltage signal that can be read and processed. This setup allows testing of the signal behavior, response consistency, and system calibration without exposing the microcontroller or operators to high currents. It is particularly useful for observing how the CT and sensor setup respond under different loads and for verifying linearity and stability of the measurements.

2.5.1.1.4 IR Temperature Sensor



Figure 15 - IR Temperature Sensor

It has an IR (Infrared) temperature sensor that indicates the surface temperature of the machine without touching it. This makes the temperature of the shaft machine in clear sight in real time, to avoid the machine from overheating so that the machine is operated safely and efficiently. Instead the IR sensor produces a normal current signal, typically with some value between 4–20mA, that cannot be read directly by a microcontroller such as the ESP32.

2.5.1.1.5 MD0662



Figure 16 - MD0662 Module

MD0662 module used with IR temperature sensor. Because The MD0662 is a signal conditioning module that converts a standard current signal, usually in the range of 4–20 mA, into a voltage signal that microcontrollers can read, typically 0–3.3 V or 0–5 V. Many industrial sensors, such as temperature, pressure, and current sensors, output a 4–20 mA current, which cannot be directly measured by devices like the ESP32.IR temperature can read current value between 4-20mA it

can’t directly input into the microcontrollers like ESP32. MD0662 convert this value as 0–3.3 V or 0–5 V.

2.5.1.1.6 K-type Thermocouple



Figure 17 - K-Type temperature sensor

This sensor uses direct contact to monitor internal temperature, in contrast to infrared sensors. It gives accurate temperature readings of the machine's vital parts to guarantee correct operation and guard against damage.

2.5.1.1.7 Thingsboard

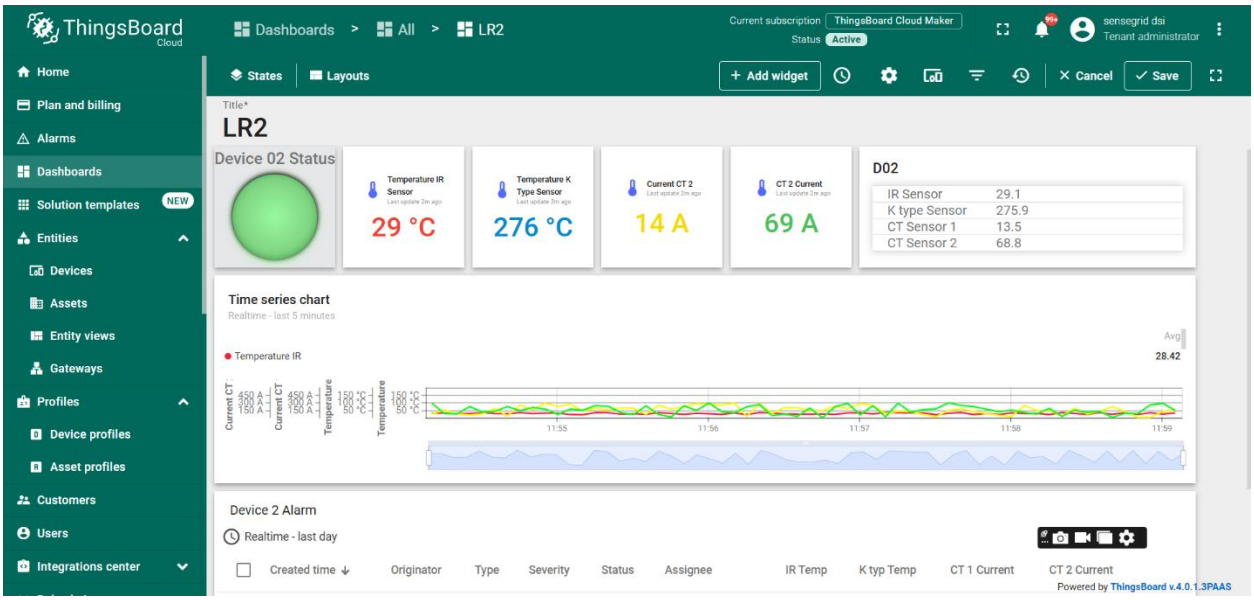


Figure 18 - Thingsboard Dashboard For DSI

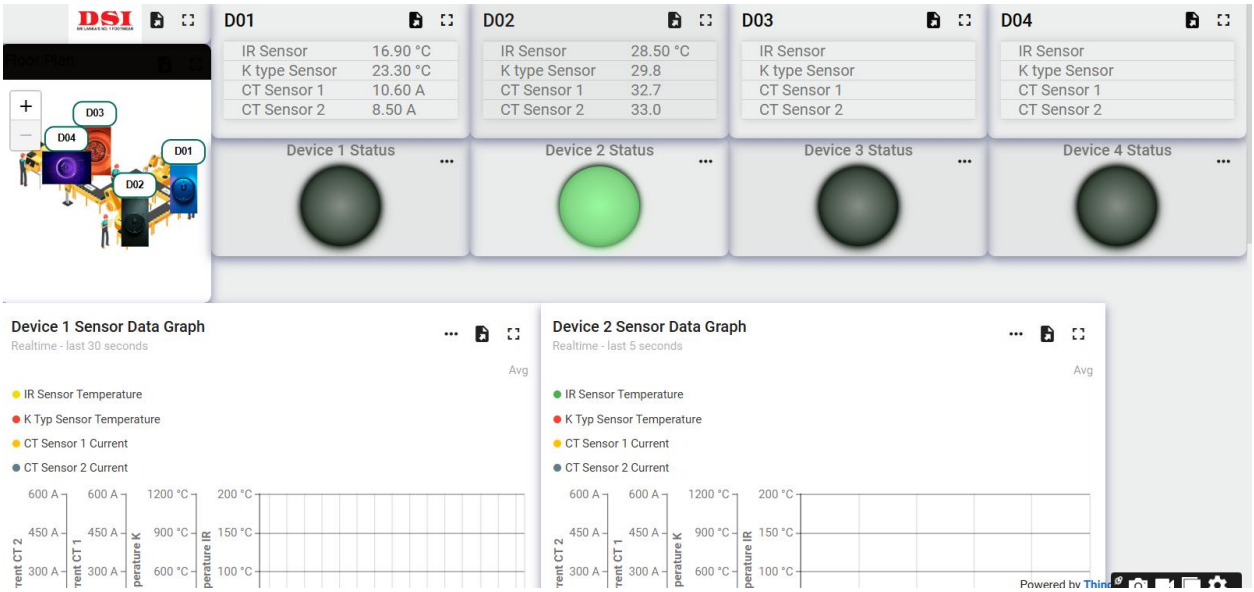


Figure 19 - DSI Dashboard For all devices

The DSI project uses five dashboards in ThingsBoard, with each device having its own dedicated dashboard. Every dashboard includes LED indicators, four value cards, an attribute card, line charts, and alarms. The LED indicators display the device status. If the green bulb is on, the device is powered on, and if it is off, the device is not active. The four-value cards display sensor readings one by one, including K-type temperature, IR, and two current transformers. These cards show both the latest sensor value and the last update time. The attribute card presents all sensor values together at once for quick monitoring. Line charts are included to visualize the sensor readings as real-time graphs, making it easy to observe changes over time. the last one is alarm widget, if sensor values are exceed threshold value alarm is triggered and d automatically sends an email notification to the customer.

All device dashboards include four LED indicators, four line charts, alarms, a value card, and an attribute card. this dashboard additionally include image map. marked image map that shows sensor locations within the factory, helping users quickly identify where each measurement is taken.

2.5.1.2 Contribution

2.5.1.2.1 Enclosure Box design

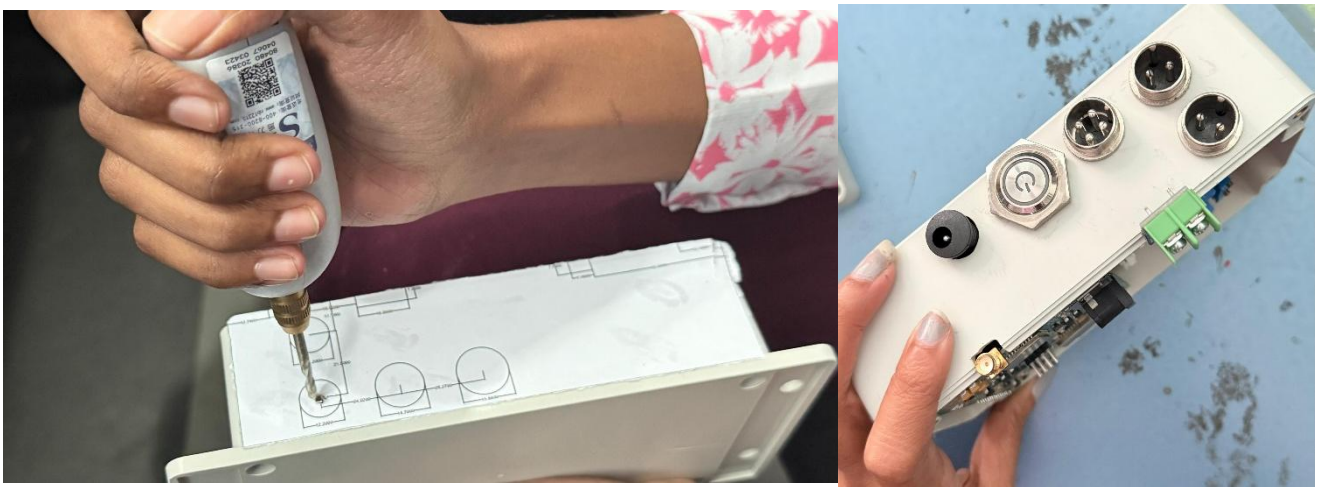


Figure 20 - Drilling for switch holes

The enclosure box was created to protect and organize all the project parts. Positions for connectors, switches, and cable entry points were marked carefully. Precision drilling made holes for sensor inputs, power cables, and output terminals, ensuring a proper fit and alignment.

Next, switches, jacks, and connectors were installed in their assigned positions. This allows for easy access to connections and control. The finished enclosure securely holds all electronics, offers durability, and ensures safe and organized system operation.

2.5.1.2.2 ADC712 Sensor Linearity Testing



Figure 21 - ADC712 Testing

The linearity and accuracy of measuring the AC current consumed by the industrial device in the DSI project were tested in terms of the ADC712-based current-to-voltage module. The objective of this test is to check whether the module output is proportional to the power load, for different power loads.

The test was performed by connecting a series of resistive loads with power capacity of 40W, 100W, 200W, 300W and 1000W and measuring the current and voltage with a digital multimeter, while the values outputted by the module were read during the entire process in the serial monitor. These measurements were performed at various load conditions to avoid diverge from possible mutant current values to the output of the module

2.5.1.2.3 Current to Voltage Module Calibrate for IR sensor

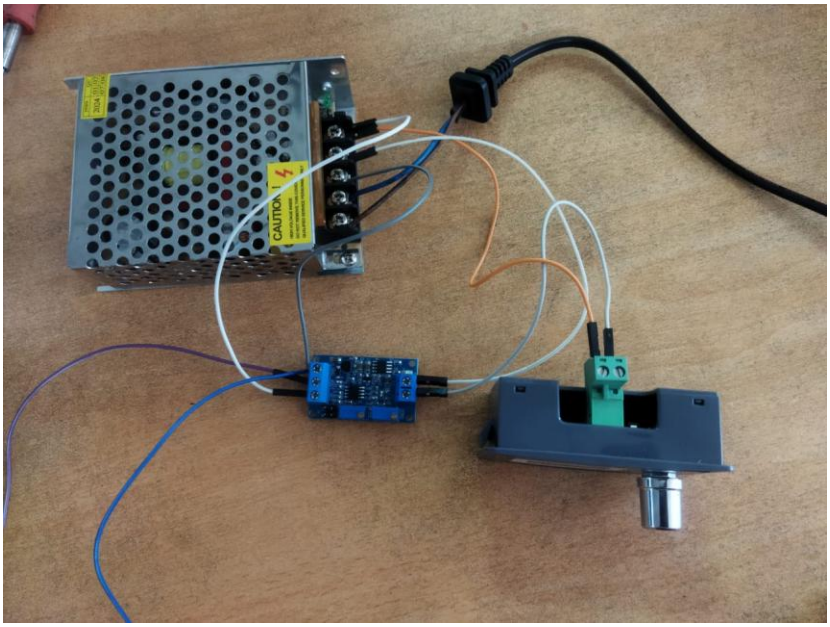


Figure 22 - Current to Voltage Module Calibrate for IR sensor

DSI project, we calibrated a module that changes the current from IR sensors into a voltage the device can read. The sensors send a current between 4 mA and 20 mA, and the module converts it to a voltage between 0 V and 3.3 V

Calibration

- 4 mA → 0 V
- 20 mA → 3.3 V

Used Component

- Another Power Supply – to simulate the sensor
- Multimeter – to measure voltage and current
- Mini Oscillator – to produce exact current levels
- DC Power Supply – to power the module

Steps

1. Connect the Module Power the module with 12 V. Connect the current source in series with the module. Use a multimeter to measure the module's output voltage.

2. Set Zero Point Send 4 mA through the module. Adjust the offset knob so the output reads 0 V.

3. Set Full Scale o Increase the current to 20 mA. Adjust the span knob so the output reads 3.3 V.

4. Check Middle Values Test currents like 8 mA, 12 mA, 16 mA. Make sure the output voltage changes smoothly and correctly.

5. Final Check - Connect the module to the ESP32 device. Make sure the device reads the correct values for different currents.

2.5.1.2.4 Thingsboard Create

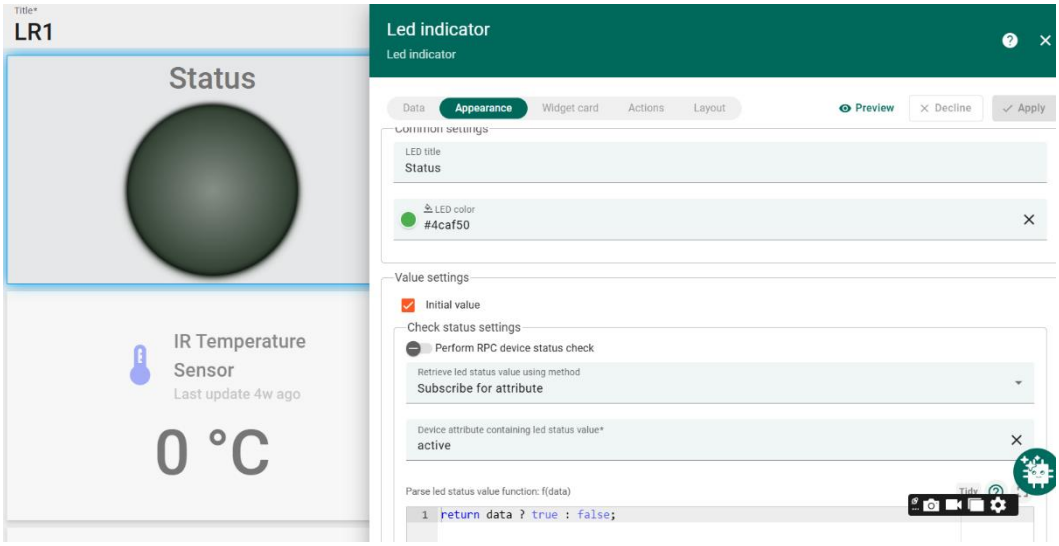


Figure 23 - LED Indicator Settings

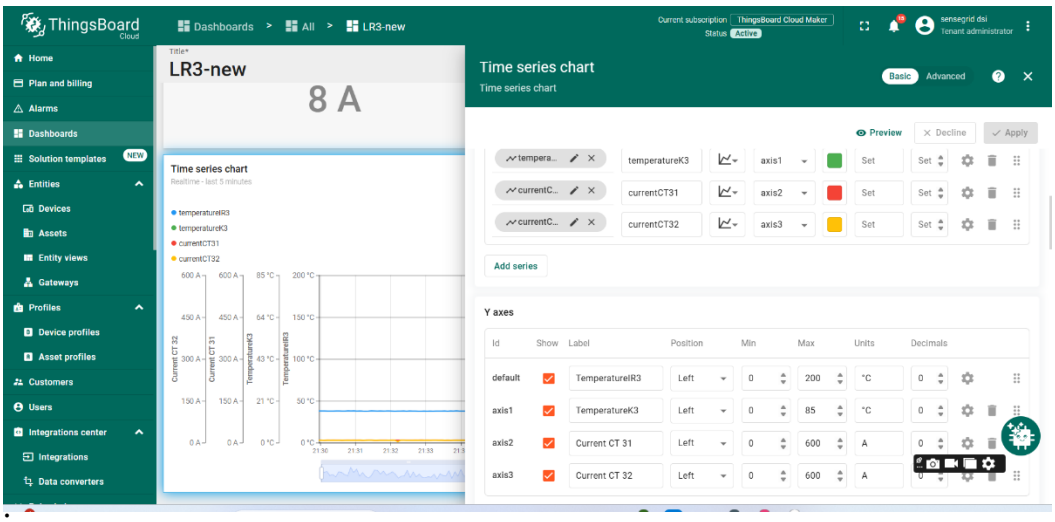


Figure 24 - Line Chart Setting

2.5.2 DHL project

In the DHL project, the main objective was to monitor the environment inside their warehouse, which used for storing medicine. Since the quality and effectiveness of medicine can affected by temperature changes, it is essential to maintain proper temperature conditions in different parts of the warehouse.

In this project, we need to create dashboard part and hardware component. Dashboard was created Using thingsboard. Thingsbard visualize real time data and data analysis. The hardware part involves sensors and devices for collecting data.

2.5.2.1 Hardware components

- ESP32 Microcontroller
- DSI8B20 digital temperature sensor
- External antenna for using strong Wi-Fi connection

2.5.2.2 Contribution

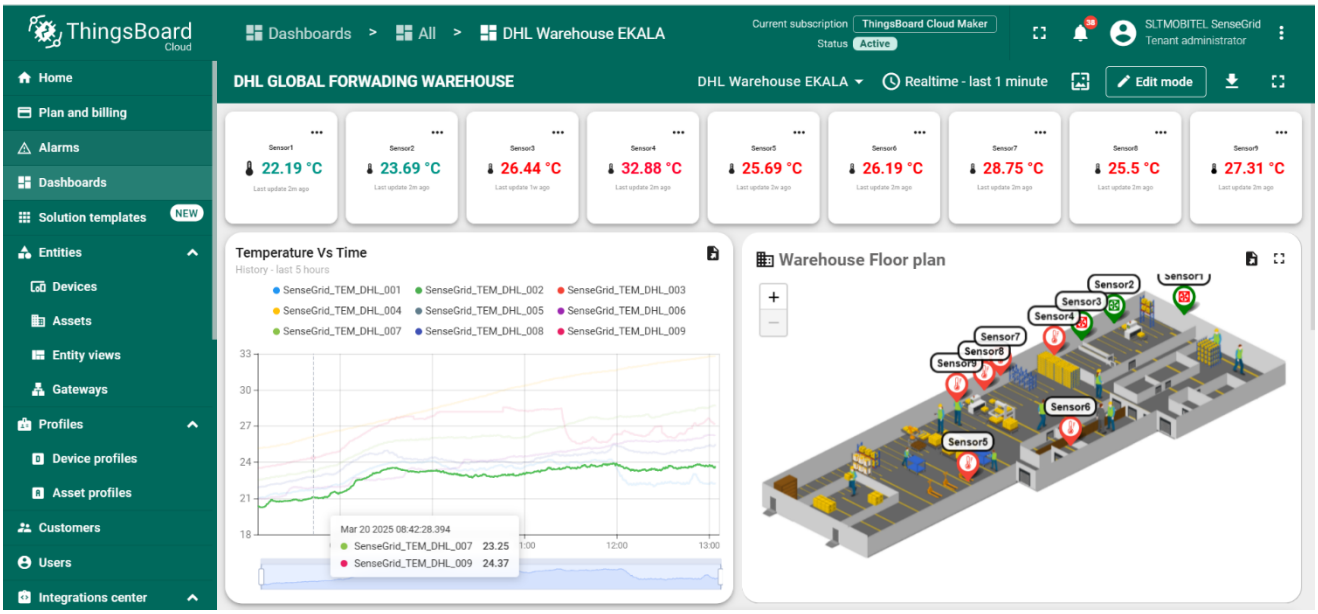


Figure 25 - DHL Dashboard

The DHL Project dashboard was edited.

- The installed device location was marked on the floor plan in things board
- Dashboard include
- Nine value card – Display Temperature Values
- Line charts – In this line chart visualize real time on 5 hours and click one line and display its value in the chart

- Floor plane – this is DHL Ekala Ware house real floor plane. Temperature marker display our device placement position and we can click this marker come to new dashboard.
- Alarms - in nine devices have 7 alarms

The alarm is set up

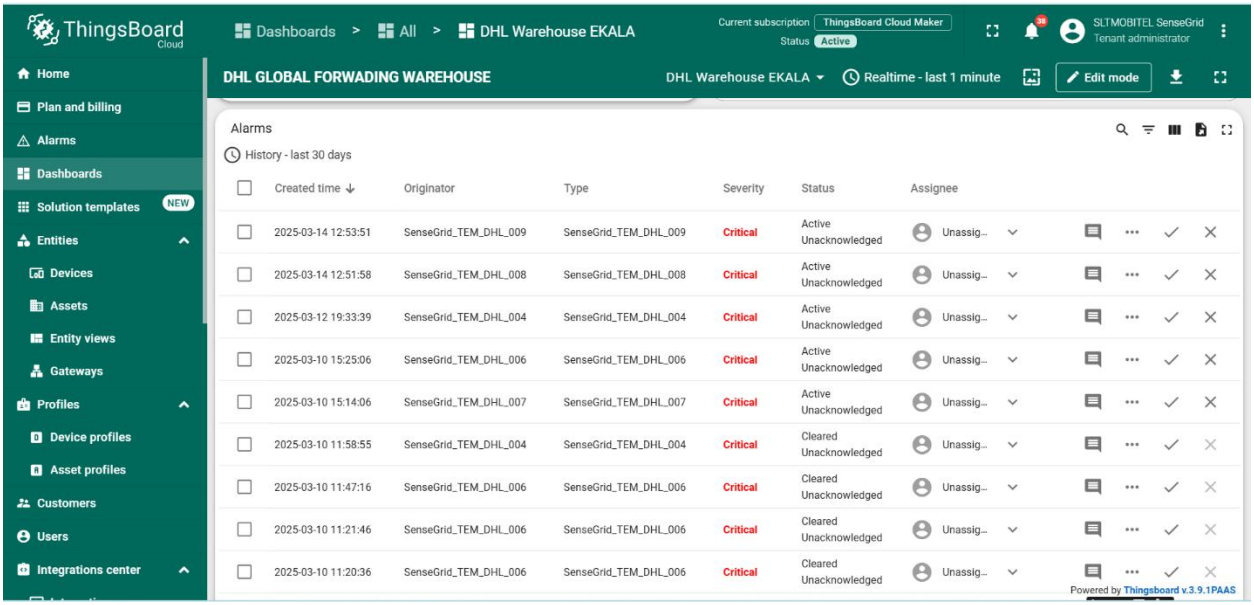


Figure 26 - DHL Alarm

The DHL project has seven devices. When the temperature value reaches the threshold, an alarm triggered. In ThingsBoard, we can create alarms in two ways: the first is by using a Rule Chain, and the second is by using the Alarm settings in the Device Profile. If we use the Rule Chain method, we need to create a separate Rule Chain for each device. This is little bit complex. We can use the device profile to add different threshold values for each device.

2.5.3 BOC Bank IoT Solution



Figure 27 - BOC project output

This project was developed for the Bank of Ceylon (BOC) to address challenges in their ATM rooms, specifically concerning continuous air conditioning, excessive power usage, and the need for a remote monitoring solution. The system's goal was to optimize environmental control and reduce energy consumption by automating AC units based on real-time temperature and occupancy data.

The main part of the system is the SLT IoT Development Board, which uses an ESP32 microcontroller as the brain of the project. This board was designed to provide strong connectivity and reliable performance.

For sensing, the system used an IR blaster, an IR receiver, and a DHT22 sensor. The DHT22 sensor measured indoor and outdoor temperature and humidity. The IR blaster allowed the system to capture and copy the AC unit's infrared signals, enabling automated control.

In terms of connectivity, the system supported both Wi-Fi for local communication and a GSM module for reliable cloud data transmission. This made the system operational even without a stable Wi-Fi connection.

The system needed a strong power supply. It used an industrial-grade AC-to-DC power supply along with a UPS battery backup, which kept the system running smoothly during power cuts.

All hardware was placed inside a custom-made enclosure that was non-conductive, fire-resistant, and well-ventilated. External connections for sensors and power were made with GX16 pins and JST connectors to ensure durability and easy modular design.

To control the AC units, the system used two types of relays. Solid-State Relays (SSRs) were used for quick and reliable switching, while Electromechanical Relays (EMRs) handled higher surge currents. This combination made the AC control safe and effective.

2.5.3.1 Contribution

I connected and tested the DHT22 sensor to make sure it could read temperature and humidity data in real time. This step was important to check the accuracy of sensor readings.

I updated the firmware so the device could handle connection issues by retrying up to 10 times and restarting if needed.

I worked on the physical fabrication by selecting the enclosure, planning and drilling holes, and soldering parts on the dot board. This made the device compact and well-organized.

2.6 Challenges and Solutions

2.6.1 AP+STA Mode

- Connectivity Issues in AP+STA Mode

  When running both AP and STA modes at the same time, the ESP32 can face unstable Wi-Fi connections. Used stable channel and Optimize Wi-Fi settings by limiting the number of AP clients

- Limited Resources on ESP32

  The ESP32 has limited processing power and memory. Running Wi-Fi in dual mode, along with MQTT and other tasks, may cause performance drops or crashes.

  Use lightweight libraries (like AsyncMqttClient)

- MQTT Message Reliability

  MQTT messages may not always arrive on time or could be duplicated. In networks with high latency or packet loss, Use QoS 2 for critical data to guarantee exactly-once delivery.

2.6.2 DSI Project

- Development board not connected to the internet. In addition, change development board antenna and its working correctly. Problem is we use mismatch frequency antenna
- Sensor Accuracy and Calibration sensor and current transformers output current signals can't directly read by ESP32. If we are not calibrate correctly sensor reading come as invalid. Using MD0662 and calibrate it using signal generator
- Continuous Data Transmission. Factory environments often face connectivity issues. These problems can interrupt data flow to ThingsBoard. The customized SLT development board included both ESP32 Wi-Fi and a 4G SIM module.
- In this project, managing large amounts of data across several dashboards was a challenge. Too much information could confuse users and delay problem detection. To address this, the dashboards included clear visual tools such as LED indicators, line charts, alarms, and image map.

2.6.3 FOTA

2.6.3.1 Challenges in ThingsBoard FOTA

- The free version has limits; advanced FOTA requires a paid plan. Use the free version for testing and the paid plan for many devices.
- ESP32 must always stay connected to Wi-Fi and MQTT. Add reconnect code for Wi-Fi and MQTT.
- Large firmware files can be difficult to download or store. Keep firmware small and ensure good Wi-Fi.

2.6.3.2 Challenges in ESP32 Web Page FOTA

- Only one device can be updated at a time. - Use Web FOTA for single device or testing, ThingsBoard FOTA for multiple devices.
- Limited authentication (simple admin/admin login is insecure).- Improve login security (custom username/password or token).
- Browser upload depends on user action, not automated. - Add checksum/MD5 verification for uploaded firmware.

2.6.4 BOC ATM Project

- We encountered a significant challenge with the IR blaster system while connecting it to a Mitsubishi AC unit. Previously, we had tested it only with LG ACs, which operate with a straightforward on/off signal. In contrast, Mitsubishi ACs use a more complex signal that includes temperature and mode in each command. To address this, we captured the exact IR codes from a Mitsubishi AC remote and incorporated them into our firmware using the IRremoteESP8266 library. As a result, our device could effectively control both Mitsubishi and other AC brands.

3 Chapter 03

3.1 Conclusion

During my internship with the Research and Development Section at Sri Lanka Telecom PLC, I gained a lot of experience in embedded systems and IoT solutions. This training provided a hands-on environment where I could connect what I learned in school with real-world applications. The organization's focus on technology allowed me to explore modern monitoring systems and IoT platforms in detail.

During my internship, I was involved in designing and developing IoT-based systems with the ESP32 microcontroller. My tasks included selecting and integrating sensors like the DHT22, IR temperature sensors, and K-type thermocouples to monitor temperature, humidity, and machine load. I worked on the system setup, configuring the ESP32 to run in both STA and AP mode for internet connectivity and direct device access. I also developed firmware for MQTT communication with ThingsBoard to enable real-time monitoring and data visualization.

Finally, I worked on ThingsBoard dashboards to visualize sensor data with value cards, line charts, LED indicators, and alarms, including email notifications when thresholds were exceeded. I implemented Firmware Over-the-Air (FOTA) updates both via ThingsBoard and web page, allowing the ESP32 to remotely check, download, and apply firmware updates, ensuring continuous and efficient system operation.

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Certification

NAME OF THE TRAINEE : Ama Hirushima Karagalage

ID NO : 200175200968

ADDRESS : 255/A, Boralugoda, Athurugiriya.

INSTITUTE & ADDRESS : Faculty of technology, University of Colombo, Mahenwatta, Pitipana – Thalagala Rd, Homagama.

DEGREE PROGRAM : Bachelor of Engineering Technology Honors in Instrumentation and Automation.

PERIOD : 4 years.

TRAINING ORGANIZATION & ADDRESS : Sri Lanka Telecom PLC, Lotus road, Colombo 01.

TRAINING PERIOD : 6 months

This is to certify that this report is prepared by myself under the training conducted at the above organization.

2025/08/21

Date

Signature of trainee

This is to certify that this report is prepared by the above trainee under my regular supervision.

RUMESH KALASEGAR B. Sc (Hons)
Research & Development Engineer
Research & Development
Sri Lanka Telecom PLC

2025/08/22

Date

Signature of trainee